

ASSESSMENT TOOL 1 -CASE STUDY

NAME : RAGAVI S

REGISTER NO : 192565027

COURSE CODE : CSA1717

COURSE NAME : ARTIFICIAL INTELLIGENCE

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis using propositional logic.

(a) Propositional Logic Representation

Symbol Definitions:

Symbol	Meaning
F	Fever
C	Cough
R	Rash
B	Breathlessness
Flu	Possible Flu
Measles	Possible Measles
Pneumonia	Possible Pneumonia

Rules:

- $R1: F \wedge C \rightarrow \text{Flu}$
- $R2: F \wedge R \rightarrow \text{Measles}$
- $R3: C \wedge B \rightarrow \text{Pneumonia}$

Facts (Patient A):

- $F = \text{True}, C = \text{True}, B = \text{True}$

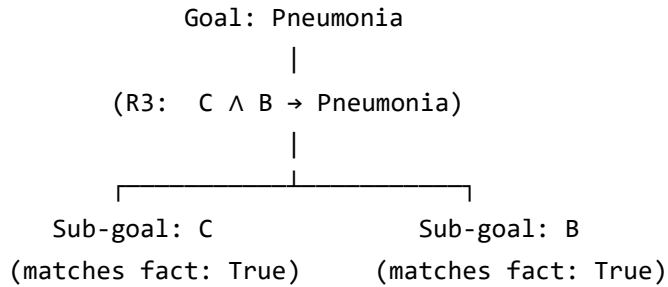
(b) Forward Chaining

Step	Facts Known	Rule Fired	New Fact Derived
1	F, C, B	$R1: F \wedge C \rightarrow \text{Flu}$	Flu
2	F, C, B, Flu	$R3: C \wedge B \rightarrow \text{Pneumonia}$	Pneumonia
3	F, C, B, Flu, Pneumonia	$R2: F \wedge R \rightarrow \text{Measles}$	Cannot fire (R unknown)

Result: Possible Flu and Possible Pneumonia are derived for Patient A. Measles is NOT derived, since Rash (R) is not present as a fact.

(c) Backward Chaining — Goal: Pneumonia

Goal reduction tree:



- Sub-goal 1: C (Cough) → matches known fact, True ✓
- Sub-goal 2: B (Breathlessness) → matches known fact, True ✓

Both sub-goals are satisfied ⇒ Goal 'Pneumonia' is CONFIRMED as True.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a First-Order Logic (FOL) knowledge base to manage emergency vehicle routing.

Rules:

- Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$
- Rule 2: $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$
- Rule 3: $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

Facts:

- $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification of Rule 1 with Facts

Rule 1: $\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

- Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance}) \rightarrow \theta_1 = \{x/\text{Ambulance}\}$
- Unify $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$ under $\theta_1 \rightarrow$ consistent, $x/\text{Ambulance}$ confirmed

Most General Unifier: $\theta = \{x / \text{Ambulance}\}$

Applying θ to Rule 1 $\rightarrow \text{ClearPath}(\text{Ambulance})$ is derived.

(b) Resolution Proof — Prove $\text{Proceed}(\text{CarA})$

CNF Conversion of Rules:

- Rule 1 (CNF): $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
- Rule 2 (CNF, relevant conjunct): $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$
- Rule 3 (CNF): $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Negated Goal: $\neg \text{Proceed}(\text{CarA})$

Resolution Steps:

1. $\text{Vehicle}(\text{Ambulance}) + \text{EmergencyType}(\text{Ambulance}) + \text{Rule1} \rightarrow \text{ClearPath}(\text{Ambulance})$
2. $\text{ClearPath}(\text{Ambulance}) + \text{Rule2} \rightarrow \text{GreenSignal}(\text{Ambulance})$
3. $\text{Rule3 } \{x/\text{CarA}, y/\text{Ambulance}\}:$
 $\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee$
 $\text{Proceed}(\text{CarA})$
4. Resolve with $\text{Vehicle}(\text{CarA}) \rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee$
 $\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
5. Resolve with $\text{Behind}(\text{CarA}, \text{Ambulance}) \rightarrow \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
6. Resolve with $\text{GreenSignal}(\text{Ambulance})$ (step 2) $\rightarrow \text{Proceed}(\text{CarA})$
7. Resolve $\text{Proceed}(\text{CarA})$ with $\neg \text{Proceed}(\text{CarA}) \rightarrow \text{EMPTY CLAUSE } (\square)$

Contradiction derived $\Rightarrow \text{Proceed}(\text{CarA})$ is PROVEN.

(c) Sufficiency for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient. Justification:

- Rule 2 assigns GreenSignal to every vehicle satisfying ClearPath, with no priority ranking.
- If two ambulances approach from conflicting directions simultaneously, both would independently satisfy Rule 1 and Rule 2, producing a logical conflict — both cannot safely hold right-of-way.

Additional rules/facts needed:

- A priority predicate: $\text{Priority}(x, n)$, with a rule such as $\text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Priority}(x) > \text{Priority}(y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$
- A mutual exclusion constraint: $\neg(\text{GreenSignal}(x) \wedge \text{GreenSignal}(y))$ for two vehicles on conflicting approaches at the same intersection
- A distance/arrival-time predicate to break ties when priorities are equal

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with a propositional knowledge base.

Rules:

- P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Facts:

- $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) CNF Conversion

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

- Eliminate implication: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
- Already in CNF ✓

P2: $(\text{IrrigationNeeded} \wedge \text{CropWheat}) \rightarrow \text{ApplyDripMethod}$

- Eliminate implication: $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$
- Move $\neg$ inward (De Morgan): $\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$
- Already in CNF ✓

P3: $(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \rightarrow \text{CropAtRisk}$

- Eliminate implication: $\neg(\neg\text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$
- Move $\neg$ inward (De Morgan): $\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$
- Already in CNF ✓

Facts (unit clauses):

- SoilDry
- CropWheat

(b) Resolution Refutation — Prove ApplyDripMethod

#	Clause	Source
1	$\neg\text{SoilDry} \vee \text{IrrigationNeeded}$	P1
2	$\neg\text{IrrigationNeeded} \vee \neg\text{CropWheat} \vee \text{ApplyDripMethod}$	P2
3	$\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$	P3
4	SoilDry	Fact
5	CropWheat	Fact
6	$\neg\text{ApplyDripMethod}$	Negated Goal

Resolution / Proof Tree:

- 7: (1,4) SoilDry + $(\neg\text{SoilDry} \vee \text{IrrN})$ $\rightarrow$ IrrigationNeeded
- 8: (7,2) IrrigationNeeded + $(\neg\text{IrrNV} \neg\text{CWWADM})$ $\rightarrow$ $\neg\text{CropWheat} \vee \text{ApplyDripMethod}$
- 9: (8,5) $(\neg\text{CropWheatVADM})$ + CropWheat $\rightarrow$ ApplyDripMethod
- 10: (9,6) ApplyDripMethod + $\neg\text{ApplyDripMethod}$ $\rightarrow$ EMPTY CLAUSE ($\square$)

Contradiction $\Rightarrow$ ApplyDripMethod is PROVEN (entailed by the KB).

(c) If 'SoilDry' is Removed

- P1 can no longer fire (it requires SoilDry as a fact) $\rightarrow$ IrrigationNeeded is NOT derivable.
- P2 can no longer fire $\rightarrow$ ApplyDripMethod is NOT derivable (its truth value becomes unknown, not proven false).
- P3 requires $\neg\text{ApplyDripMethod} \wedge \text{CropWheat}$ to derive CropAtRisk. Since ApplyDripMethod is unknown (not proven false), strict resolution CANNOT derive CropAtRisk.
- Under a Closed-World Assumption / Negation-as-Failure (common in expert systems), anything not proven is treated as false $\rightarrow$ $\neg\text{ApplyDripMethod}$ assumed $\rightarrow$ combined with CropWheat, CropAtRisk would follow as a default inference.

Conclusion: Under classical (open-world, sound) resolution, CropAtRisk does NOT follow. It only follows as a non-monotonic / default inference under the Closed-World Assumption — not as a valid logical proof.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2].

Rules:

- Rule 1: Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- Rule 2: Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- Rule 3: Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- Rule 4: $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Belief State — Modus Ponens Applications

Step	Modus Ponens Application	Derived Conclusion
1	Stench[1,2] is True; Rule 1: Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]	Wumpus adjacent to [1,2]: candidates {[1,1],[1,3],[2,2]}
2	Breeze[1,1] is True; Rule 2: Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]	Pit adjacent to [1,1]: candidates {[1,2],[2,1]}
3	Glitter[2,2] is True; Rule 3: Glitter[x,y] $\rightarrow$ Gold at [x,y]	Gold is at [2,2]

Derived facts: Wumpus $\in$ {[1,1],[1,3],[2,2]}, Pit $\in$ {[1,2],[2,1]}, Gold = [2,2]

(b) Forward Chaining — Locating Wumpus and Pit

Step	Trigger	Inference
1	Stench[1,2]	Wumpus adjacent to [1,2] $\rightarrow$ candidates: [1,1],[1,3],[2,2]
2	Agent started safely at [1,1] (safe by convention)	Eliminate [1,1] $\rightarrow$ Wumpus $\in$ {[1,3],[2,2]}
3	Breeze[1,1]	Pit adjacent to [1,1] $\rightarrow$ candidates: [1,2],[2,1]
4	No Breeze perceived at [1,2]	Pit $\neq$ [1,2] $\rightarrow$ Pit = [2,1]
5	Pit = [2,1]; Gold (not Wumpus) confirmed at [2,2]	Wumpus = [1,3] (by elimination)

Conclusion: Pit at [2,1], Wumpus at [1,3], Gold at [2,2].

(c) Safety of Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

Using Rule 4: $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

- [1,1]: Starting cell, safe by convention. ✓
- [1,2]: Wumpus $\neq$ [1,2] (eliminated in part b), Pit $\neq$ [1,2] (no breeze perceived) $\rightarrow$ $\neg$ Wumpus[1,2] $\wedge$ $\neg$ Pit[1,2] $\rightarrow$ Safe[1,2]. ✓
- [2,2]: Wumpus $\neq$ [2,2] (deduced as [1,3]), Pit $\neq$ [2,2] (Pit = [2,1]) $\rightarrow$ $\neg$ Wumpus[2,2] $\wedge$ $\neg$ Pit[2,2] $\rightarrow$ Safe[2,2]. Also confirmed by Glitter $\rightarrow$ Gold, not a hazard. ✓

Conclusion: The path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is logically SAFE for the agent to traverse and collect the gold, verified using Modus Ponens and Forward Chaining on Rule 4.